

# Descriptive Qwen and NVIDIA Remote-Model Panels for Fixed-Policy LLM Play in Four-Player Go-Stop

Go-Stop AI Evaluation Harness  
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## Abstract

We evaluate fixed-policy large language model play in four-player Go-Stop, a Korean card game with stochastic deals, seat-order nuisance effects, multi-agent interaction, and risk-sensitive payoffs. The study asks whether observed performance across a Qwen-labeled remote-model panel is monotone in nominal parameter scale, and whether cross-family differences are visible in a rate-limited, roughly same-scale NVIDIA Build panel.

The Qwen panel compares four DashScope models spanning 30B, 122B, 397B, and 480B total parameters. The NVIDIA Build panel compares four roughly 100B-122B models from Qwen, Mistral, Nemotron, and Stockmark. Each panel yields 48 session samples per model and 192 model-session observations. The results are descriptive, not universal model rankings.

## 1. Introduction

This paper is about evaluation, not training. We study fixed language-model policies in four-player Go-Stop, where agents choose among legal game actions under partial information, stochastic deals, seat-order effects, and heavy-tailed settlements.

The motivating claim is simple: model scale is an incomplete proxy for game-playing policy quality. A multi-agent, risk-sensitive game can expose differences in action selection, risk tolerance, and instruction following that are not captured by parameter count alone.

RQ1: In a Qwen-labeled DashScope panel, is observed Go-Stop performance monotone in nominal parameter scale?

RQ2: In a rate-limited NVIDIA Build panel of roughly same-scale models, are cross-family differences visible in descriptive profit and risk summaries?

## 2. Methods

The harness evaluates fixed policies only. There is no online learning, self-play training, or log-based policy update. Each agent receives a public game-state serialization and must return one legal action index.

Remote models are called through OpenAI-compatible chat-completion APIs. The fixed policy prompt asks the model to choose one legal action index only, optimize long-run profit with risk control, and avoid explanation. Invalid, unparsable, or out-of-range responses fall back to the first legal action.

The Qwen study uses Alibaba DashScope with all 24 seat permutations repeated twice. Each session has a two-hand horizon. The NVIDIA study uses NVIDIA Build NIM with the same seat enumeration and repetition, but a one-hand horizon and `max_remote_calls_per_agent=1` because rate limits were binding.

The primary metric is terminal session profit relative to initial bankroll. We also report empirical CVaR<sub>0.05</sub>, win rate, ruin probability, and sample count. Each model has 48 session samples, so the effective 5% tail mass is 2.4 sessions and CVaR is descriptive.

### 3. Results

RQ1 is descriptively negative in this panel. The highest Qwen result is the 397B model, not the largest 480B model. Because the panel mixes Qwen3.5 and Qwen Coder variants, it does not isolate a pure scale effect.

| Model                          | Total params | Active params | Mean profit | CVaR 5%  | Win rate |
|--------------------------------|--------------|---------------|-------------|----------|----------|
| qwen3.5-397b-a17b              | 397B         | 17B           | 996.25      | -1850.08 | 0.3021   |
| qwen3.5-122b-a10b              | 122B         | 10B           | -51.75      | -4145.33 | 0.3542   |
| qwen3-coder-480b-a35b-instruct | 480B         | 35B           | -198.92     | -5084.75 | 0.2396   |
| qwen3-coder-30b-a3b-instruct   | 30B          | 3B            | -745.58     | -5269.25 | 0.1042   |

RQ2 shows visible descriptive differences under the constrained NVIDIA panel. Nemotron has the highest mean profit and least severe empirical CVaR among the roughly same-scale models.

| Model                               | Scale | Mean profit | CVaR 5%  | Win rate |
|-------------------------------------|-------|-------------|----------|----------|
| nvidia/nemotron-3-super-120b-a12b   | ~120B | 500.65      | -1384.17 | 0.2708   |
| mistralai/mistral-small-4-119b-2603 | ~119B | -73.19      | -4150.00 | 0.2917   |
| qwen/qwen3.5-122b-a10b              | 122B  | -172.85     | -3350.42 | 0.1875   |
| stockmark/stockmark-2-100b-instruct | ~100B | -254.60     | -4267.08 | 0.2500   |

### 4. Discussion

The two panels agree on a bounded point: fixed-policy Go-Stop performance in this harness is not fully summarized by nominal parameter count. In the Qwen panel, the highest descriptive result comes from the 397B model rather than the 480B model. In the NVIDIA panel, roughly same-scale models separate in descriptive profit and empirical tail summaries, with Nemotron ranking first.

These outcomes fit the game domain. Go-Stop requires local tactical choices, risk control, settlement awareness, and disciplined JSON action selection. General language-model scale can help, but the observed policy is also shaped by model specialization, instruction following, decoding behavior, provider-level serving constraints, and fallback behavior.

### 5. Limitations

The model panels are convenience samples from currently callable remote APIs. The Qwen scale panel mixes Qwen3.5 and Qwen Coder variants, so parameter scale is confounded with model specialization.

The NVIDIA panel is constrained by rate limits and uses max\_remote\_calls\_per\_agent=1. It measures constrained first-decision-policy behavior more than full-game autonomous play. The horizons are short: two hands for Qwen and one hand for NVIDIA.

Each model has 48 session samples, below the repository's 100-session threshold for stable CVaR inference. The exported tables do not report invalid-response, API-error, cache-hit, or fallback-action rates. The sample count is enough to compute descriptive summaries, but not enough for universal model ranking claims.

### 6. Conclusion

In a reproducible four-player Go-Stop evaluation harness, the observed Qwen panel is not monotone in nominal parameter scale, and the constrained NVIDIA Build panel shows visible cross-family differences in descriptive profit and empirical tail summaries. The highest Qwen result comes from qwen3.5-397b-a17b; the highest NVIDIA result comes

from nvidia/nemotron-3-super-120b-a12b.

The main implication is methodological: multi-agent, risk-sensitive card-game evaluation should report model identity, parameter scale, family, serving constraints, fallback policy, and tail-risk metrics together rather than reducing model quality to size alone.

## Code and Artifact Availability

Repository: <https://github.com/parktaejun-dev/ai-test-with-gostop>

Web paper page: <https://godori.dahanda.dev/dashboard/paper.html>

Manuscript source: `paper/arxiv_main.tex`

Qwen result bundle: `results/paper_qwen_4model_param_2h_2rep`

NVIDIA result bundle: `results/paper_nvidia_120b_4model_family_1h_2rep_budget1`

Compact result summary: `paper/remote_model_results.md`

## Reproducibility Commands

```
python3 scripts/run_dashscope_qwen_parameter_sweep.py \  
--session-hands 2 \  
--layout-repetitions 2 \  
--remote-eval-hands 2 \  
--decoding max_tokens=64 \  
--decoding temperature=0 \  
--decoding enable_thinking=false \  
--output-dir results/paper_qwen_4model_param_2h_2rep
```

```
python3 scripts/run_nvidia_same_size_family_eval.py \  
--session-hands 1 \  
--layout-repetitions 2 \  
--remote-eval-hands 1 \  
--max-remote-calls-per-agent 1 \  
--decoding max_tokens=64 \  
--decoding temperature=0 \  
--output-dir results/paper_nvidia_120b_4model_family_1h_2rep_budget1
```

References are maintained in `paper/refs.bib`. The TeX manuscript remains the canonical source for citation keys and bibliography style.